

Log-Derivative Trick in Optimization

The log-derivative trick, also known as the REINFORCE trick or score function estimator, is a technique used in stochastic optimization, particularly in the context of machine learning and reinforcement learning. It is used to compute gradients of expected values with respect to parameters when the expectation is difficult to handle due to the involvement of stochastic (random) variables.

Key Idea

Suppose you have an objective function that involves an expectation with a probability distribution parameterized by θ :

$$J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [f(x)]$$

where $p(x; \theta)$ is a probability distribution over x , and $f(x)$ is some function of x .

The goal is to compute the gradient of $J(\theta)$ with respect to the parameter θ . Directly differentiating under the expectation can be challenging if x is sampled from $p(x; \theta)$.

Log-Derivative Trick

The log-derivative trick allows us to express this gradient in a form that simplifies the computation:

1. Rewrite the gradient of the expectation:

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{x \sim p(x; \theta)} [f(x)]$$

Using the fact that the gradient of an expectation can be moved inside the expectation (a detailed explanation for this step is available in the bottom of this document):

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [\nabla_{\theta} f(x) + f(x) \nabla_{\theta} \log p(x; \theta)]$$

If $f(x)$ does not depend on θ , the first term $\nabla_{\theta} f(x)$ drops out, leaving us with:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [f(x) \nabla_{\theta} \log p(x; \theta)]$$

2. Estimate the gradient using samples:

Since the expectation can be approximated using Monte Carlo sampling, we sample $x_1, x_2, \dots, x_n$ from $p(x; \theta)$ and compute the gradient as:

$$\nabla_{\theta} J(\theta) \approx \frac{1}{n} \sum_{i=1}^n f(x_i) \nabla_{\theta} \log p(x_i; \theta)$$

By using samples, the log-derivative trick enables gradient-based optimization methods even when the distribution $p(x; \theta)$ is complex.

Applications

- **Reinforcement Learning:** In policy gradient methods, where the policy is parameterized by θ , the log-derivative trick is used to derive the gradient of expected rewards with respect to policy parameters.
- **Variational Inference:** In variational autoencoders and other probabilistic models, it helps in optimizing parameters of distributions.

The log-derivative trick is a powerful method because it turns a challenging problem (differentiating expectations) into an easier one that can be directly tackled using samples and gradient estimates.

Supplementary

The transformation from $\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E}_{x \sim p(x; \theta)} [f(x)]$ to $\nabla_{\theta} J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [\nabla_{\theta} f(x) + f(x) \nabla_{\theta} \log p(x; \theta)]$ is based on the application of the differentiation under the integral sign, also known as the "differentiation of an expectation" or the Leibniz rule for differentiation under the integral sign. Here is a detailed explanation:

Gradient of an Expectation

Given an expectation with respect to a parameterized probability distribution $p(x; \theta)$:

$$J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [f(x)] = \int p(x; \theta) f(x) dx,$$

To find the gradient with respect to θ , we differentiate under the integral sign:

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \int p(x; \theta) f(x) dx.$$

Using the Product Rule

Applying the product rule of differentiation to the integrand $p(x; \theta) f(x)$, we get:

$$\nabla_{\theta} (p(x; \theta) f(x)) = f(x) \nabla_{\theta} p(x; \theta) + p(x; \theta) \nabla_{\theta} f(x).$$

Note that $\nabla_{\theta} f(x)$ is zero when $f(x)$ does not depend on θ ; otherwise, it is included in the gradient.

Integral of the Gradient

$$\nabla_{\theta} \int p(x; \theta) f(x) dx = \int [f(x) \nabla_{\theta} p(x; \theta) + p(x; \theta) \nabla_{\theta} f(x)] dx.$$

Using the Log-Derivative Trick

We utilize the log-derivative trick to simplify $\nabla_{\theta} p(x; \theta)$. The trick is based on the identity:

$$\nabla_{\theta} p(x; \theta) = p(x; \theta) \nabla_{\theta} \log p(x; \theta).$$

Substituting this into the integral, we get:

$$\int [f(x)p(x; \theta)\nabla_{\theta} \log p(x; \theta) + p(x; \theta)\nabla_{\theta} f(x)] dx.$$

This can be rewritten as:

$$\int p(x; \theta) [f(x)\nabla_{\theta} \log p(x; \theta) + \nabla_{\theta} f(x)] dx.$$

Moving the Gradient Inside the Expectation

Recognizing that the integrated function represents the expectation with respect to $p(x; \theta)$:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [f(x)\nabla_{\theta} \log p(x; \theta) + \nabla_{\theta} f(x)].$$

Simplification

If $f(x)$ does not depend on θ , then $\nabla_{\theta} f(x) = 0$, and the final expression simplifies to:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{x \sim p(x; \theta)} [f(x)\nabla_{\theta} \log p(x; \theta)].$$